

YASH RAJ

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SUMMARY

AI/ML Engineer with production experience in LLM fine-tuning, autonomous agent systems, and high-throughput inference optimization. Built and shipped enterprise-grade AI systems achieving 30–35% latency reduction and 90%+ task accuracy. Expertise in open-source LLM ecosystems (vLLM, llama.cpp, Hugging Face), multi-tool agent architectures, RAG pipelines, and scalable MLOps.

TECHNICAL SKILLS

Core ML / AI: PyTorch · TensorFlow · Hugging Face Transformers · Scikit-learn · OpenCV · Multimodal Learning

LLM & GenAI: LLM Fine-tuning (LoRA, QLoRA, PEFT) · RAG Systems · Autonomous AI Agents · Tool Calling · Vision-Language Models

Inference & Optimization: vLLM · Ollama · llama.cpp · ONNX · TensorRT · Quantization · Pruning

Vector & Retrieval: FAISS · ChromaDB · Semantic Search · Embedding Models · Hybrid Retrieval

MLOps & Cloud: MLflow · DVC · Docker · GitHub Actions · AWS (EC2, S3) · Azure Container Apps · Apache Spark

Languages & APIs: Python · SQL · C++ · JavaScript · FastAPI · REST API

Data & Analysis: Pandas · NumPy · Matplotlib · EDA · Feature Engineering · Statistical Modeling

EXPERIENCE

Software Engineer – AI/ML | ProSkep Solutions LLP

Noida, UP · Jul 2025 – Present

- Fine-tuned open-source LLMs on large-scale enterprise datasets using **QLoRA/PEFT**, achieving **90%+ accuracy** on domain-specific benchmarks; improved response relevance for business workflows.
- Architected the core AI engine of **FS Suite** — an enterprise platform for intelligent document processing, knowledge extraction, and workflow automation using LLM-based pipelines.
- Built and deployed scalable LLM inference pipelines on AWS EC2, achieving **30–35% reduction in P99 latency** for real-time workloads via continuous batching and quantization.
- Benchmarked **10+ open-source transformer LLMs** across accuracy, throughput, and cost to select optimal models for domain-specific deployment.
- Developed production REST APIs in FastAPI/Python; validated under high-concurrency load testing for stable, scalable AI service delivery.

Software Development Engineer | AST Consulting

Noida, UP · Nov 2024 – Mar 2025

- Built a scalable HRMS platform on Frappe/ERPNext, reducing operational costs by **60%** through end-to-end workflow automation.
- Optimized MariaDB/MySQL relational schemas, improving query performance by **30%**; automated attendance, leave, and hiring workflows, increasing HR team efficiency by 50%.

ML Research Intern | Ignitus Labs

Mexico (Remote) · Feb 2024 – Aug 2024

- Engineered a multimodal deep learning pipeline integrating Sentinel imagery, Landsat cubes, and tabular data for plant species distribution analysis across 20+ European countries and 10,000+ data points.
- Achieved **top baseline performance** on the GeoLifeCLEF benchmark using a Siamese network variant combining ResNet18, Swin-v2-t, and feed-forward fusion with AdamW + BCE loss.
- Improved predictive accuracy by **20%** via systematic hyperparameter tuning; outputs used for conservation planning and biodiversity management.

PROJECTS

Excel AI Agent — Autonomous Financial Auditing & Document Intelligence · Python · CrewAI · FastAPI · openpyxl · RAG · Vectorless RAG · ReAct · 2025 – Present

- Engineered a **production-grade autonomous AI agent** for financial auditing using a **ReAct-style tool-calling architecture** capable of multi-step reasoning over Excel workbooks, structured financial documents, and live web sources.
- Built **agent tools**: Excel read/write, formula execution, financial Q&A (RAG via LlamaIndex/ChromaDB), audit report generation, real-time policy lookup via web search, and external API integration (currency, tax).
- Implemented a **document grounding pipeline** that retrieves live regulatory/policy documents from the web, cross-references them against uploaded financials, and flags discrepancies automatically.
- Designed write-back capability allowing the agent to directly modify Excel cells, inject formulas, and generate structured audit summary sheets from natural language instructions.

GeoStitch — Drone Footage to GeoTIFF Map Overlay · Python · OpenCV · Rasterio · Folium · NumPy · May 2025

- Built an end-to-end pipeline converting georeferenced drone video into stitched GeoTIFF maps using **SIFT feature matching**, RANSAC homography, and GCP-based georeferencing (WGS84/EPSG:4326).
- Deployed Folium-based overlay visualization providing high-fidelity spatial analysis for AI training datasets and geographic modeling.

Multimodal Species Distribution Modeling · PyTorch · ResNet18 · Swin-v2-t · CUDA · Scikit-learn · Jun 2024

- Implemented a fusion architecture combining CNN, Vision Transformer, and feed-forward networks for multi-source ecological data; addressed class imbalance and sparse label challenges at scale.
- Achieved **0.3163 validation loss (top baseline on GeoLifeCLEF benchmark track)** — enabling groundwork for real-time ecological monitoring.

AWARDS & ACHIEVEMENTS

- **UGC NET JRF (2024)** — Computer Science & Applications · **98.17 percentile**
- **1st Place** — University-Level Web application Development Competition, Bytecode Learners (2022)

EDUCATION

B.Tech – Computer Science & Engineering

Central University of Haryana, Mahendragarh, HR

Oct 2020 – Aug 2024